Technology Stack

Date	15 July 2026
Team ID	SWTID-2026-5339
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Used to build a responsive, user-friendly interface for students and administrators that works across different devices.
2	Backend / Server-Side	Python, Flask	Flask provides a lightweight framework for handling user authentication, AI integration, course management, and application logic efficiently.
3	Database / Data Storage	MySQL	Stores user accounts, course details, quiz results, learning progress, and recommendation data securely with efficient relational database management.
4	Cloud / Hosting / Deployment	Render / PythonAnywhere	Provides reliable hosting, easy deployment, scalability, and continuous availability of the application.
5	Version Control & CI/CD	Git, GitHub	Enables source code management, version tracking, collaboration among team members, and backup of project code.
6	Third-Party APIs / Other Tools	Google Gemini API, VS Code, GitHub	Gemini API powers AI-based learning recommendations and chatbot features, while VS Code is used for development and GitHub for project collaboration and repository management.